



PES University, Bengaluru

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024

UE20CS351 - Cloud Computing

Total Marks : 100.0

1.a. Discuss most common technology challenges that occur in cloud computing
(8.0 Marks)

1.b. Describe briefly 3 service models and 3 deployment models of cloud computing.
(6.0 Marks)

1.c. Explain the four core concepts of publisher/subscriber model used in asynchronous communication. What are the different purposes for which the pub/sub model can be used.
(6.0 Marks)

2.a. Explain Docker architecture with a neat diagram (5 Marks) . How does a Docker container differ from a VM? (3 Marks) (8.0 Marks)

2.b. Highlight the key differences between Type 1 Hypervisor and Type 2 Hypervisor? (6.0 Marks)

2.c. What are the challenges of shadow page table in a VMM? How does Nested/Extended page table overcome the software overheads of shadow paging? (6.0 Marks)

3.a. Explain Lustre file system architecture with a neat diagram. (7.0 Marks)

3.b. Briefly explain CAP theorem? (2 M) Explain how would you choose a database for your application based on CAP theorem (3M) (5.0 Marks)

3.c. What is replication in a distributed environment? Explain leaderless replication. (5.0 Marks)

3.d. What is rebalancing of partitions? Why is it necessary to rebalance partitions? (3.0 Marks)

4.a. Explain Ring Election Algorithm with neat sketches. Clearly state the worst-case scenario and messages required in worst case scenario. (6M)
What are the changes made in Modified Ring Election Algorithm to address the problem in Ring Election Algorithm? (2M) (8.0 Marks)

4.b. Analyse the best case scenario and the worst case scenario of Bully Algorithm. (6.0 Marks)

4.c. What is Zookeeper? How does it work? Name at least two contexts where zookeeper services may be used. (6.0 Marks)

5.a. Explain the following types of design pattern:
1. CMDB
2. Defense in Depth
3. Honeypots
4. Network Patterns (8.0 Marks)

5.b. Discuss the Cloud Scaling Strategies. (4M) How does Cloud Scalability differ from Cloud Elasticity? (2M) (6.0 Marks)

5.c. What are the benefits of reverse proxy? How is it different from forward proxy?
Provide a few applications where these proxies are used. (2M + 2M + 2M)
(6.0 Marks)